

Subject Description Form

Subject Code	EE520
Subject Title	Intelligent Motion Systems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To describe an in depth knowledge on the design and operation of intelligent motion systems. 2. To relate and compare numerous application examples, which ranges from CD players and hard disc drives to robots and component insertion machines. 3. To enable the students to have the ability to design motion control systems for industry and domestic purposes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Contrast and compare different motion control system configurations, and select the most appropriate one for the task. To comprehend and understand numerous motion control examples for domestic and industrial applications. b. Understand the in-depth knowledge of motion drive and sensing techniques, and the ability to use them in real engineering applications. c. Have a broad understanding of motion control platform hardware and a visionary perspective on the future developments of computing/control hardware.
Subject Synopsis/ Indicative Syllabus	1. Structures of intelligent motion systems: Specifications and requirements of intelligent motion systems. Operating modes: point to point motion, trajectory path tracking, velocity path tracking, force and tension control, compliance control, vibration damping. Switching between operation modes. 2. Motion actuators and driving techniques: Using Voice Coil Motors and DC brush motors in motion control. AC brushless motors, linear direct drive AC brushless motors and their driving techniques. Stepping motors and their limitations in motion tracking systems. Microstepping and electronic damping of stepping motors. 3. Motion sensing and estimation techniques: Optical encoders: working principle, decoding method, and resolution enhancement through interpolation. Syncro-resolvers: working principle and interface electronics. Velocity estimation and position estimation methods for large speed range actuators. 4. Motion control platform: Computer hardware requirements. Tightly coupled systems versus distributed systems. Application of DSPs in motion control. Communication methods in motion systems. Real time operating system for motion control. 5. Intelligent algorithms for motion control and trajectory generation: PID controllers and their variations. Servo tuning methods. Motion control systems based on state space configuration. States observation and Kalman filters. Using Notch filters in non-rigid systems. Profile generation and motion planning algorithms. 6. Issues in multi-axis intelligent motion systems: co-ordinate mapping and dynamics transformation. Multi-axis motion planning and profile generation. Motion synchronisation between axis. Decoupling inter-axis motion interference. Applying MIMO structure in tightly coupled system.

	7. Case studies in intelligent motion systems: Three examples will be selected from the following list: a. Optical based position tracking in CD-ROMs and Laser discs. b. Magnetic head positioning in hard disk drives. c. Motion control system design in multi-axis robot manipulators. d. Gantry robot motion systems for SMT component insertion machines. e. Motion systems in high precision CNC tooling machines. Case study: Report on a high performance motion control application example																																				
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures, complemented by tutorials and worked examples. Self-learning on the part of students is strongly encouraged and extensive use of web resources will be made. A term paper and a related presentation enable students to develop skills in literature survey and writing. Oral presentation sessions develop students’ skills in spoken communication and peer evaluation. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>Lectures</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorials</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Assignment and oral presentation</td><td>√</td><td>√</td><td>√</td></tr></table>				Teaching/Learning Methodology	Outcomes			a	b	c	Lectures	√	√	√	Tutorials	√	√	√	Assignment and oral presentation	√	√	√														
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Test</td><td>30%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Report</td><td>5%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Oral presentation</td><td>5%</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td></tr></table> One end-of-semester written examination; one mid-semester-test; one end-of-semester test; a report on an assigned topic; and a PowerPoint presentation for the particular topic.				Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Examination	60%	√	√	√	2. Test	30%	√	√	√	3. Report	5%	√	√	√	4. Oral presentation	5%	√	√	√	Total	100%			
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Student Study Effort Expected	Class contact:																																				
	▪ Lecture/Tutorial		30 Hrs.																																		
	▪ Presentation/Test		9 Hrs.																																		
	Other student study effort:																																				
	▪ Case study		18 Hrs.																																		
	▪ Self-study		48 Hrs.																																		
	Total student study effort		105 Hrs.																																		
Reading List and References	References books: 1. Hakan Gurocak, Industrial Motion Control: Motor Selection, Drives, Controller Tuning, Applications, Wiley, 2016 2. Terry L.M. Bartelt, Industrial Automated Systems: Instrumentation and Motion Control, DELMAR, Cengage Learning, 2011 3. Lynch, K.M. and Park, F.C., Modern Robotics: Mechanics, Planning, and Control, Cambridge University Press, 2017																																				